Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

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Coordinates on a Manifold

We will be interested in systems on manifolds

$$\dot{\xi} = f(\xi, u)$$

How do we describe $\xi \in \mathcal{M}$ for a general system on a manifold?
In classical differential geometry texts, coordinates on a manifold are given by an atlas of charts

$$\mathcal{A} = \{\alpha_i : U_i \to \mathbb{R}^m \,|\, U_i \subset \mathcal{M}, i \in \mathcal{I}\}$$

where U_i are open subsets in $\mathcal{M}$, α_i are diffeomorphisms, and $\mathcal{I}$ is a countable index set.

Equivariant systems theory provides a different way to describe points on a manifold.

Let $W_i \subset \mathbb{R}^n$ be the range of α_i. Since α_i is a diffeomorphism then W_i is open and $\alpha_i^{-1} : W_i \to U_i$ is itself a diffeomorphism. α_i^{-1} is called a local *parametrisation*.

Symmetry parametrisation

STEP 1: Let $\phi_X : \mathcal{M} \to \mathcal{M}$ for $X \in \mathbf{G}$ be a transitive family of symmetries on $\mathcal{M}$

- Transitive: for any $\xi_1, \xi_2 \in \mathcal{M}$ there exists $X \in \mathbf{G}$ such that $\phi_X(\xi_1) = \xi_2$.
- **G** is just an index set set - but will be a Lie-group later in the course!

STEP 2: Choose an arbitrary $\mathring{\xi} \in \mathcal{M}$. Call this point the *origin*.

STEP 3: Define $\phi_{\mathring{\xi}} : \mathbf{G} \to \mathcal{M}$ by

$$\phi_{\mathring{\xi}}(X) = \phi_X(\mathring{\xi})$$

Then **G** is a global surjection of **G** onto $\mathcal{M}$.

STEP 4: A point $\xi \in \mathcal{M}$ is represented by an element $X \in \mathbf{G}$ such that $\xi = \phi_{\mathring{\xi}}(X)$

The set **G** can be used as a sort of (high dimensional) *global* parametrisation of $\mathcal{M}$.

Symmetry parametrisation

Why is this a good thing to do

- The parametrization is global
- We can use ϕ to define global *error* between $\phi_{\mathring{\xi}}(X_1) = \xi_1$ and $\phi_{\mathring{\xi}}(X_2) = \xi_2$
$$e(\xi_1, \xi_2) = \phi_{X_1} \phi_{X_2}^{-1}(\mathring{\xi})$$
- If **G** is a Lie-group then we have a very deep understanding of coordinates, curvature, and structure of **G**.

Although the general framework $\phi_{\mathring{\xi}} : \mathbf{G} \to \mathcal{M}$ is good perspective on what equivariant systems theory is - we are much better starting with a concrete example.

Parametrising pose of a robot using $\mathrm{SE}(3)$ is an example of using symmetry to parameterise state.

Outline

1. Perspective

2. Physical Variables
   - Understanding Pose
   - Coordinates: A physical perspective

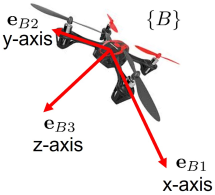
3. Rigid-body Symmetry $\mathrm{SE}(3)$
   - Rigid Transformation of Euclidean space
   - The Special Euclidean Group $\mathrm{SE}(3)$

4. Group Coordinates
   - Coordinates — An equivariant view
   - Body versus Spatial Error

Body fixed frame

Classical models of physical systems live in Euclidean three-space. The goal of this lecture is to revise the standard language and model constructions for these systems from the perspective of equivariant systems.

The state of a rigid-body is represented by the pose of a *body-fixed frame* $\{B\}$ attached rigidly to the physical component of the system.

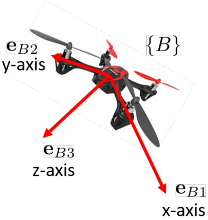


We term this:

- ⋆ The body-fixed frame (BFF).
- ⋆ The moving frame.
- ⋆ The target frame.

Physical Observer

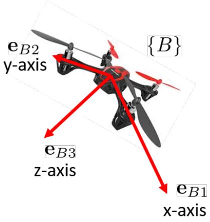
Note that talking about the pose (or any state) of a physical object is meaningless without introducing the concept of *reference*.



The physics literature uses the term *physical observer* to denote the entity or process that **measures** or **senses** or **experiences** the physical world. An *ideal physical observer* is one that does not effect the physical world by sensing it, something that is not true in quantum physics but is (mostly) true for everything we will do.

Reference origin

Because we are in the game of designing *state observers* the terminology *physical observer* is a poor choice.

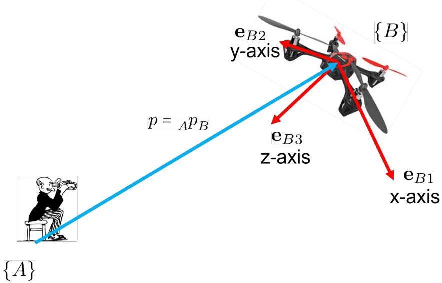


We will talk about the *reference origin* or just *origin*.

For pose space the reference origin will be a reference frame.

Position

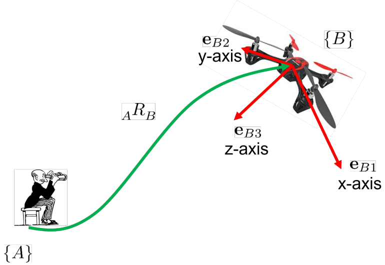
The position of the body-fixed frame is its location in the reference frame



We write $p = {}_A p_B$ to denote the abstract physical position of $\{B\}$ w.r.t. $\{A\}$.

Orientation or Attitude

The orientation of a body-fixed frame with respect to a reference frame is given by *observing* the relative directions of the BFF principle directions.

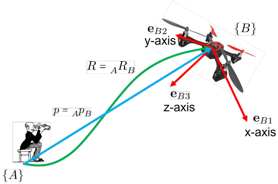


The relative orientation or attitude of a body-fixed frame is an abstract physical variable.

$$ {}_AR_B \equiv ({}_A\mathbf{e}_{B1}, {}_A\mathbf{e}_{B2}, {}_A\mathbf{e}_{B3}) $$

Pose

The pose of the body-fixed frame is simply its orientation and its position.



The pose is the pair of physical variables

$$ {}_AX_B = ({}_AR_B, {}_Ap_B) $$

Note that pose is always a relative concept; that is, ${}_AX_B$ is the pose of $\{B\}$ relative to $\{A\}$.

Coordinates

The physical variables discussed above are abstract variables in the sense that they are not expressed in coordinates. One may reason about the variables, they have physical meaning, however, you cannot realise a calculation (code them into a computer) until you express them in coordinates.

Coordinates for physical variables in pose space can be derived from physical intuition.

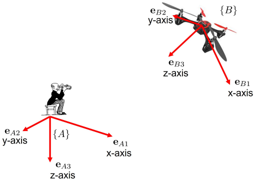
Coordinates for more abstract states like {the set of images of a planar scene} do not have a simple expression in terms of a frame of reference!

We will first revise the classical reference frame coordinates for pose - think of these analogous to the atlas of local coordinates for a general manifold.

Then we will provide a construction to show how to build coordinates for pose from the $\mathrm{SE}(3)$ pose symmetry.

Reference frame

In the case of Euclidean space the perspective of the *physical observer* is captured in the concept of *reference frame*.

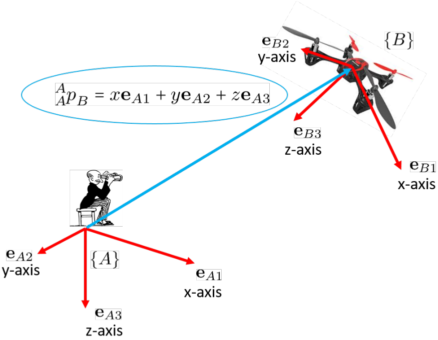


Think of the reference frame as the *point of view* through which the world is **experienced**.

For pose space we will use the term reference frame since it is standard in the literature. We use the term reference origin or just origin for more abstract spaces.

Coordinates: Classical approach

Coordinates of a point



The measurements x (resp. y, z) is the distance in the $\mathbf{e}_{A1}$ (resp. $\mathbf{e}_{A2}$, $\mathbf{e}_{A3}$) direction as observed from the observer reference frame!

We write $p = {}^A_A p_B = (x, y, z)$ to denote the coordinates of $\{B\}$ w.r.t. $\{A\}$ expressed in $\{A\}$.

Coordinates for Orientation

Writing the principle axes of the body-fixed frame with respect to the coordinates of the reference frame fully specfies (indeed over specifies) the orientation of $\{B\}$ with respect to $\{A\}$.

$$R := {}^A_A R_B = \begin{pmatrix} {}^A_A\mathbf{e}_{B1} & {}^A_A\mathbf{e}_{B2} & {}^A_A\mathbf{e}_{B3} \end{pmatrix} \in \mathbb{R}^{3\times 3}$$

This is the standard rotation matrix representation of orientation where the columns of R are ortho-normal vectors defining a right-handed triple. The matrix rotation coordinates

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} \mid R^\top R = I_3, \det(R) = 1\}$$

are associated with the matrix realisation of the Special Orthogonal group $\mathrm{SO}(3)$, however, ...

For reasons that will become clear in this lecture, it is a bad idea to think of attitude as a group element.

Coordinates for Pose

Coordinates for pose are just the concatenation of coordinates for position and orientation.

$$X = {}^A_A X_B = \left({}^A_A R_B, {}^A_A p_B\right) \in \mathbb{R}^{3\times 4}$$

Another way of writing this down is as a homogeneous matrix

$$X = {}^A_A X_B = \begin{pmatrix} {}^A_A R_B & {}^A_A p_B \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 4}$$

Both are just stacks of numbers that you code in your computer. We will use either of these representations as is convenient.

Coordinates for Pose

The set of all poses is an example of a frame bundle.

1. The set of all points in Euclidean 3-space is denoted $\mathbb{R}^3$.
2. The tangent bundle $T\mathbb{R}^3$ is the collection

$$T\mathbb{R}^3 = \{(x, \mathbb{R}^3) \mid x \in \mathbb{R}^3\}$$

3. Noting that $T_x\mathbb{R}^3 \equiv \mathbb{R}^3$ then the frame bundle is

$$\mathcal{F}_{T\mathbb{R}^3} = \{(x, {}_A\mathbf{e}_1, {}_A\mathbf{e}_2, {}_A\mathbf{e}_3) \mid ({}_A\mathbf{e}_1, {}_A\mathbf{e}_2, {}_A\mathbf{e}_3) \in \mathcal{F}_{T_x\mathbb{R}^3}\}.$$

4. A pose $({}^A_Ap_B, {}^A_AR_B)$ written in reference frame coordinates corresponds to a frame bundle element

$$x = {}^A_Ap_B, {}^A_A\mathbf{e}_1 = {}^A_AR_B\mathbf{e}_1, {}^A_A\mathbf{e}_2 = {}^A_AR_B\mathbf{e}_2, {}^A_A\mathbf{e}_3 = {}^A_AR_B\mathbf{e}_3.$$

Key Concepts

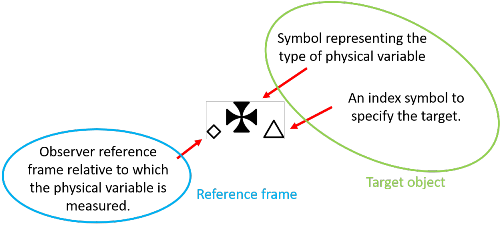
There are three key concepts that I have just introduced:

1. The idea of relativity and of a physical variable.
2. The idea of reference frame that formalises the perspective of a physical observer for pose space and will act as the origin of the relative pose representation.
3. The idea of coordinates — as a separate concept from the abstract physical variable.

For a general manifold, the origin will be a point $\mathring{\xi} \in \mathcal{M}$ and the coordinates will be an atlas $\mathcal{A}$.

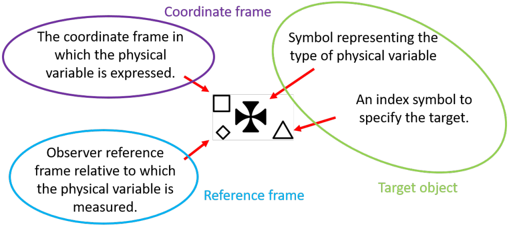
Note that we need more than just the atlas! Why is this?

Notation



The physical variable

The coordinate expression for the physical variable



In practice, we mostly use variables without superscripts or subscripts and rely on context to avoid "*les débouchés d'indices*".

Outline

Rigid transformations 1

Rigid transformation of space is a separate concept from representing the pose of a rigid-body.

Rigid transformation of space is mapping $X : \mathbb{R}^3 \to \mathbb{R}^3$ such that

$$\begin{aligned} ||X(p_1) - X(p_2)||_2 &= ||p_1 - p_2||_2 \\ (X(p_1) - X(q))^\top (X(p_2) - X(q)) &= (p_1 - q)^\top (p_2 - q), \\ &\text{for all } p_1, p_2, q \in \mathbb{R}^3 \end{aligned}$$

That is a map that preserves relative distances and relative angles!

Rigid transformations parametrization

Given a rigid-transformation $X : \mathbb{R}^3 \to \mathbb{R}^3$ we want to be able to write down how the rigid-body transformation acts

In order to express the action of a rigid-transformation on physical variables it is necessary to express the physical variables in physical **coordinates**.

The rigid-transformation can then be expressed in terms of some algebraic **parametrization**.

The action of the rigid-transformation on the physical variable can then be written as an **algebraic function** of the parametrization and the coordinates, expressed in terms of coordinates.

Rigid transformations are not physical variables.

Rigid transformations 2

Lemma (Euler): For any rigid transformation of space $X : \mathbb{R}^3 \to \mathbb{R}^3$ there exists $Q \in \mathrm{SO}(3)$ and $q \in \mathbb{R}^3$ such that

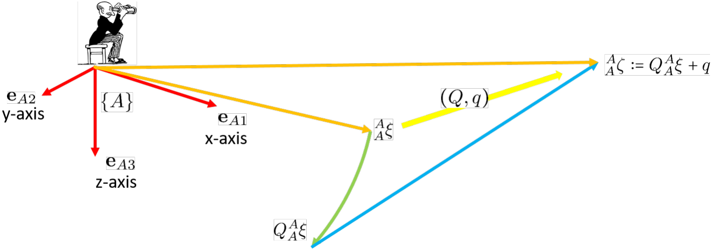
$$X(p) = Qp + q$$

Exercise: Prove this result using only the axioms of rigid-transformation.

Rigid transformation (Q, q) in reference frame coordinates.

Consider the reference frame $\{A\}$ and use coordinates $\{A\}$.

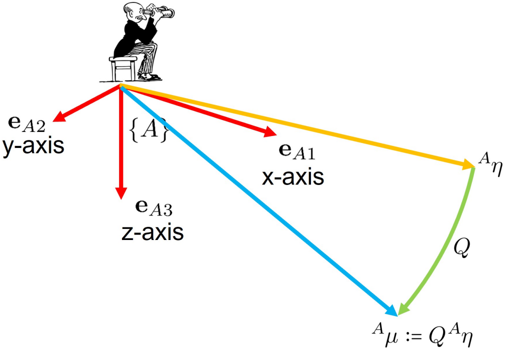
Then $\xi = {}^A\xi$ maps to a new point $\zeta = {}^A\zeta$



The matrix Q and vector q are just parameters that encode the transformation !

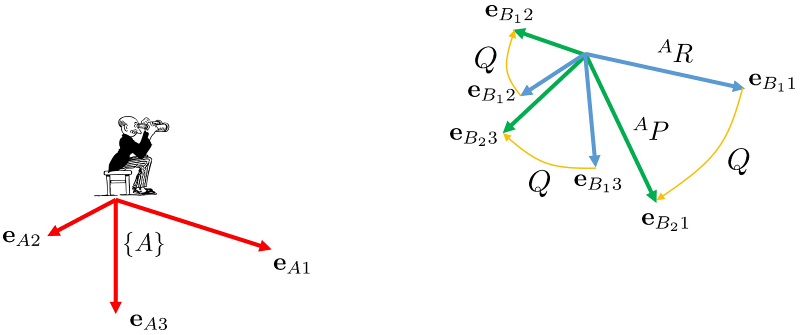
Rigid transformation directions

What about rigid transformation of directions $\eta \in \mathrm{S}^2$
Then $\eta = {}^A\eta$ maps to a new direction $\mu = {}^A\mu$



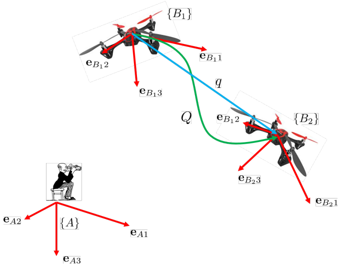
Rigid transformation orientation

This extends to a rotation of a frame $R = ({}_A\mathbf{e}_{B1}, {}_A\mathbf{e}_{B2}, {}_A\mathbf{e}_{B3})$.
Simply apply the rotation to each coordinate vector



$$^A_A R_2 = Q\, ^A_A R_1$$

Rigid transformation of a frame



The same is true for a frame

$$
\begin{aligned}
{}^A_A p_{B_2} &= Q\, {}^A_A p_{B_1} + q \\
{}^A_A R_{B_2} &= Q\, {}^A_A R_{B_1}
\end{aligned}
$$

We say that the frame $\{\mathcal{B}_1\} \mapsto \{\mathcal{B}_2\}$ transforms under the rigid-body transformation (Q, q).

We will write

$$\{\mathcal{B}_2\} = (Q, q)\{\mathcal{B}_1\}$$

for the *action* of $(Q, q) \in \mathrm{SE}(3)$ on $\{\mathcal{B}_1\} \in \mathcal{F}_{T\mathbb{R}^3}$

Homogeneous transformation matrices

Let $({}^A_A p_{B_1}, {}^A_A R_{B_1})$ and $({}^A_A p_{B_2}, {}^A_A R_{B_2}) \in \mathcal{F}_{T\mathbb{R}^3}$ be the coordinates for the two poses.

Then

$$\begin{aligned} {}^A_A p_{B_2} &= Q\, {}^A_A p_{B_1} + q \\ {}^A_A R_{B_2} &= Q\, {}^A_A R_{B_1} \end{aligned}$$

Homogeneous matrix coordinates provide a convenient algebra to compute this identity using matrix multiplication

$$\begin{pmatrix} {}^A_A R_{B_2} & {}^A_A p_{B_2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} Q & q \\ 0 & 1 \end{pmatrix} \begin{pmatrix} {}^A_A R_{B_1} & {}^A_A p_{B_1} \\ 0 & 1 \end{pmatrix}$$

The group of rigid transformations

A **group** is a set $\mathbf{G}$ of elements $X, Y \ldots \in \mathbf{G}$ with an associative multiplication $XY \in \mathbf{G}$, inverse $X^{-1} \in \mathbf{G}$ and identity $\mathrm{id} \in \mathbf{G}$ such that

$$X^{-1}X = \mathrm{id} = XX^{-1} \qquad \mathrm{id}X = X = X\mathrm{id}$$

The set of rigid body transformations is a group

$$\begin{aligned} (Q, q)(P, p) &= (QP, Qp + q) \\ (Q, q)^{-1} &= (Q^\top, -Q^\top q) \\ \mathrm{id} &= (I_3, 0) \end{aligned}$$

This group is known as the Special Euclidean Group $\mathrm{SE}(3)$.

Matrix Realisation of $\mathrm{SE}(3)$

A matrix realisation of a group is a one-to-one correspondence of group elements with real $N \times N$ matrices such that

1. group multiplication corresponds to matrix multiplication
2. group inverse is matrix inverse
3. the group identity corresponds to the identity matrix.

The Special Euclidean group has a matrix realisation as homogeneous matrices

$$(Q, q) \mapsto \begin{pmatrix} Q & q \\ 0 & 1 \end{pmatrix}$$

Conclusion

In this section we learnt that

1. Rigid body symmetry of Euclidean space is also a symmetry of pose space.
2. The special Euclidean group $\mathrm{SE}(3)$ is the group of rigid-body symmetries and has a matrix realisation as homogeneous matrices.

Note that the family of maps $L_{(Q,q)} : \mathcal{F}_{T\mathbb{R}^3} \to \mathcal{F}_{T\mathbb{R}^3}$

$$L_{(Q,q)}(R, p) = (QR, Qp + q)$$

for $(Q, q) \in \mathrm{SE}(3)$ is a transitive family of symmetries for pose space.

Note that L here is not acting as a rigid-body transformation of Euclidean space!

Outline

Group coordinates for $\mathrm{SE}(3)$

Elements of $\mathrm{SE}(3)$ are not poses!

But $\mathrm{SE}(3)$ is a manifold $\mathrm{SE}(3)$ and one can still talk of (local) coordinates for elements of $\mathrm{SE}(3)$.

The natural coordinates are the matrix elements (Q, q) of the matrix representation of a rigid-body transformation.

There is no reference frame or physical interpretation for these coordinates—they are just a parametrization of elements of the group $\mathrm{SE}(3)$.

Rigid transformations are not physical variables.

Coordinatizing pose using group coordinates

Beware! The following construction underlies the major confusion in pose representation

The group coordinates can be used to place **group** coordinates on the frame bundle of poses!

The idea

1. Fix a reference frame $\{A\}$ (origin)
2. For any pose $\{B\}$, find the group element $({}_AQ_B, {}_Aq_B)$ that transforms $\{A\}$ to $\{B\}$

$$\{B\} = ({}_AQ_B, {}_Aq_B) \cdot \{A\}$$

3. Let $({}_AQ_B, {}_Aq_B)$ denote the group coordinates of $\{B\}$ with respect to $\{A\}$.

Note: This construction is not specific to $\mathrm{SE}(3)$ and poses, it is general to any group acting (freely and transitively) on any manifold. Homographies acting on images is an example!

Rigid transformation of the origin

Lets consider the **special case** where we consider a rigid transformation $X = (Q, q) \in \mathrm{SE}(3)$ that takes the origin $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, 0_3)$ of $\{A\}$ (expressed in $\{A\}$) to a given point (R, p) in space.

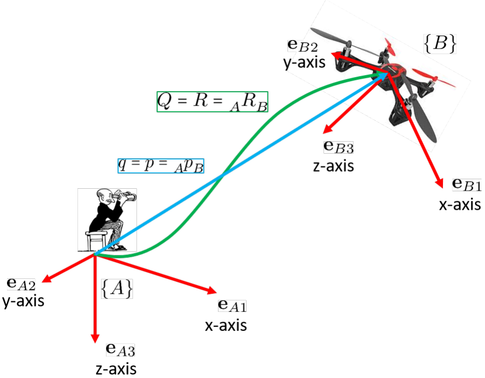
$$Q\mathbf{e}_1 = {}^{A}\mathbf{e}_{B1}, \quad Q\mathbf{e}_2 = {}^{A}\mathbf{e}_{B2}, \quad Q\mathbf{e}_3 = {}^{A}\mathbf{e}_{B3}$$
$$X(0) = Q0 + q = {}^{A}_{A}p_B$$

It is easily verified that

$$Q = R = {}^{A}_{A}R_B, \qquad q = p = {}^{A}_{A}p_B$$

That is, the matrix pair (R, p) associated with the physical coordinates of a body-fixed frame $\{B\}$ are the same as the matrix pair (Q, q) associated with the matrix coordinates of rigid transformation that takes the **origin** of $\{A\}$ to $\{B\}$ expressed in the reference frame $\{A\}$.

Group Coordinates



Physical Coordinates
$({}^A_A R_B, {}^A_A p_B)$

Group Coordinates
$({}_A Q_B, {}_A q_B)$

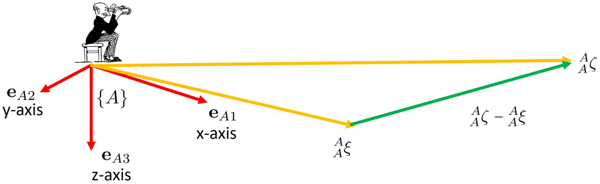
$$
{}^A_A R_B = {}_A Q_B
$$
$$
{}_A p_B = {}_A q_B
$$

Be careful! Confounding Rigid body transformations with frame coordinates is at the heart of many many many conceptual issues in robotics.

Difference of position

Once you have a group then it is possible to start defining the difference between poses. This is critically important in observer design to obtain an error between the observer state and the system state.

Consider the question of computing the difference between two points ξ and ζ. Expressing both points in the same coordinates and subtracting yields



This is possible for position since the state lies in a **linear vector space**.

The same idea is not possible for more general state objects such as rotation or pose since those state spaces are not linear.

Rigid-body transformation as a difference operator

For two poses it is possible to define a "difference" as an element of the group of rigid-body transformations.

Let $E(\{B\}, \{A\})$ denote the difference of $\{B\}$ from $\{A\}$. We cant write "$-$" since linear subtraction is ill defined.

Define

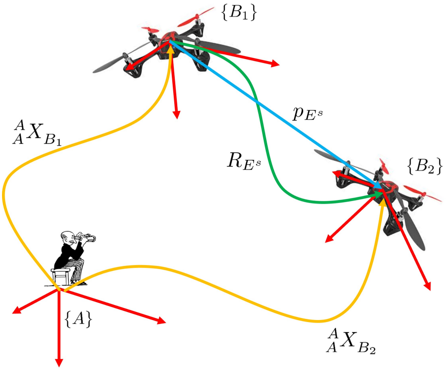
$$E(\{B\}, \{A\}) = (Q, q) \in \mathrm{SE}(3)$$

such that $(Q, q)\{A\} = \{B\}$.

Note that the difference or *error* operator

$$E : \mathcal{F}_{T\mathbb{R}^3} \times \mathcal{F}_{T\mathbb{R}^3} \to \mathrm{SE}(3)$$

Difference of poses: Spatial Error

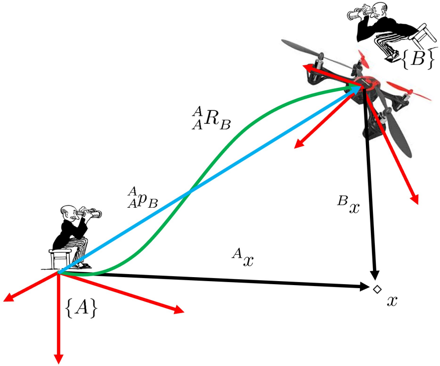


$$^A_A X_{B_1} = \begin{pmatrix} R_1 & x_1 \\ 0 & 1 \end{pmatrix}$$

$$^A_A X_{B_2} = \begin{pmatrix} R_2 & x_2 \\ 0 & 1 \end{pmatrix}$$

Given coordinates $^A_A X_{B_1}$ for frame $\{\mathcal{B}_1\}$ and $^A_A X_{B_2}$ for frame $\{\mathcal{B}_2\}$, compute a matrix representation for the rigid body transformation $E^s = (R_{E^s}, p_{E^s})$

Change of coordinates



$$^A x = {}^A R_B \, {}^B x + {}^A p_B$$

$$^A\overline{x} = \begin{pmatrix} ^A R_B & ^A p_B \\ 0 & 1 \end{pmatrix} {}^B\overline{x}$$

$$^A\overline{x} = \begin{pmatrix} ^A x \\ 1 \end{pmatrix} \qquad {}^B\overline{x} = \begin{pmatrix} ^B x \\ 1 \end{pmatrix}$$

The coordinates of $\mathbb{R}^3$ with respect to the reference $\{B\}$ are changed to coordinates of $\mathbb{R}^3$ with respect to $\{A\}$.

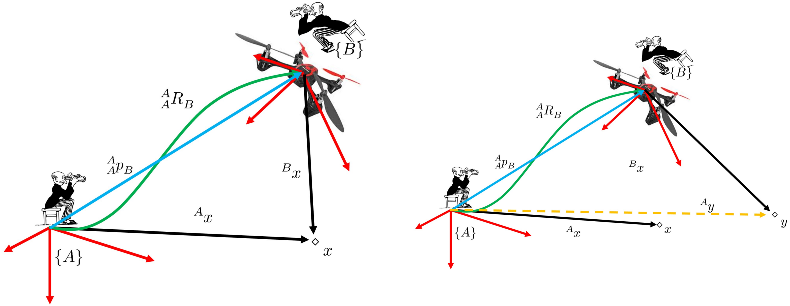
Change of Coordinates versus Transformation

Change of coordinates

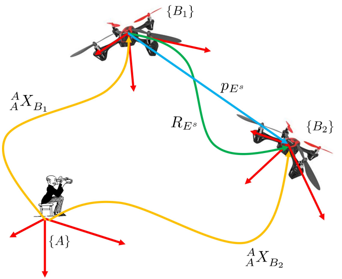
$$ {}^A x = {}^A R_B \, {}^B x + {}^A p_B $$

Physical Transformation of space

$$ {}^A y = {}^A R_B \, {}^A x + {}^A p_B $$

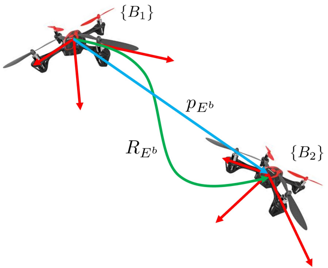


Body Error versus Spatial Error



The spatial error is the physical transformation that takes $\{B_1\}$ to $\{B_2\}$ in coordinates $\{A\}$

$$E^s = {}^A_A X_{B_2} {}^A_A X_{B_1}^{-1}$$



The body error is the physical transformation that takes $\{B_1\}$ to $\{B_2\}$ in coordinates $\{B_1\}$

$$\begin{aligned} E^b &= {}^A_A X_{B_1}^{-1} \left({}^A_A X_{B_2} {}^A_A X_{B_1}^{-1} \right) {}^A_A X_{B_1} \\ &= {}^A_A X_{B_1}^{-1} {}^A_A X_{B_2} \end{aligned}$$

Key points about rigid transformations

- The concept of rigid transformation is defined independently of coordinates. It is a property of a certain type of mapping of Euclidean space.
- Any rigid body transformation is equivalent to a rotation and translation of Euclidean space. As such it also acts to transform one pose to another.
- Group coordinates can be used to coordinatize pose by finding the group element that transforms a reference frame to the target frame.
- Group action can be used to define a “difference” or **error** between poses.
- The spatial error and the body error are different objects.